

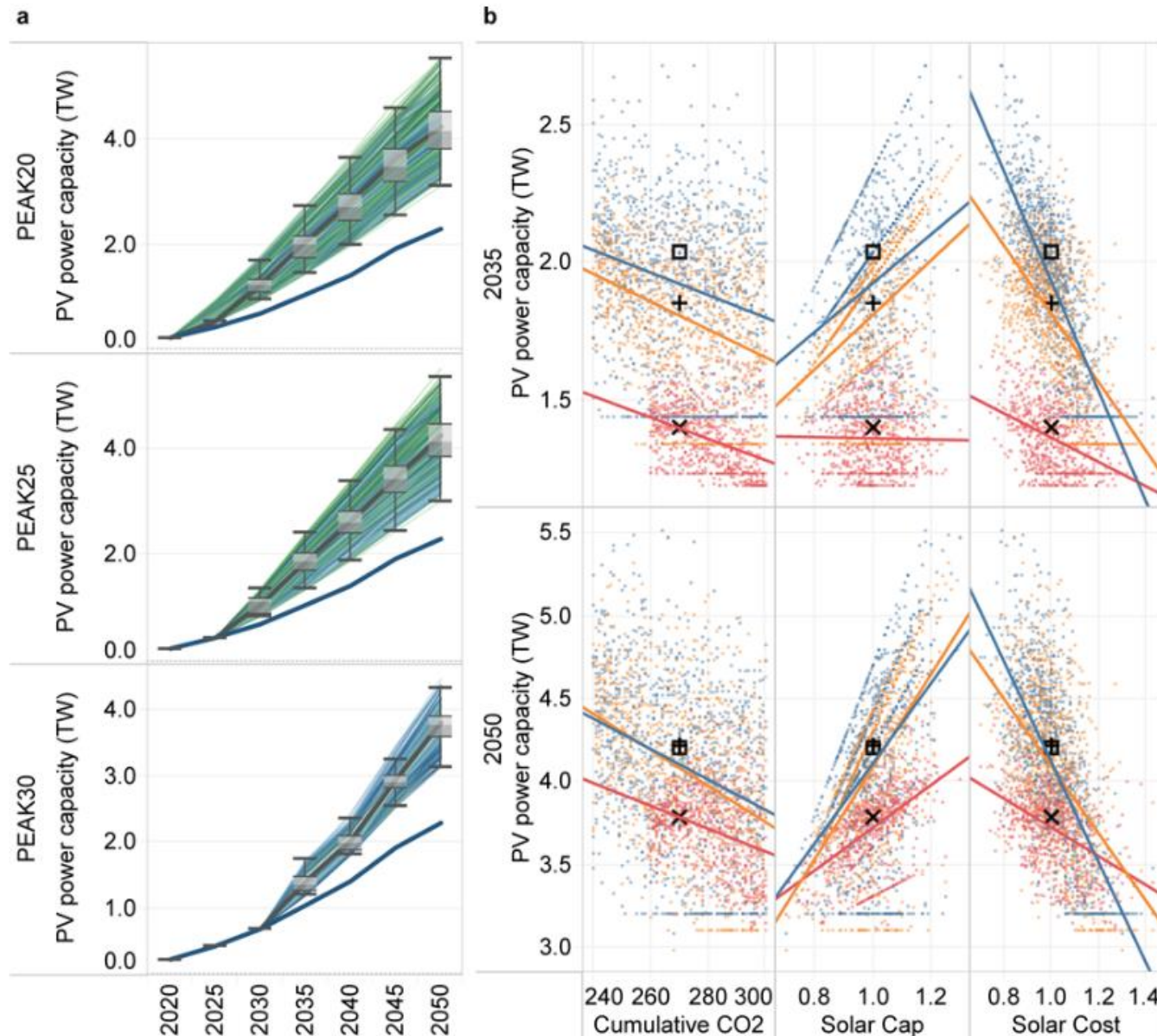
GCAM-HPC-PKU

基于计算集群的GCAM工具

Jerry Song 宋京阳
jerrysong2025@stu.pku.edu.cn
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Modelling scenario ensemble



A large scenario ensemble is needed to capture multidimensional **uncertainties**, quantify the range and likelihood of possible transition pathways, and identify **robust strategies** across diverse futures.

Socio-economy (SSPs)

middle-of-the-road; taking-the-green-road
a-rocky-road; a-road-divided; taking-the-highway



Radiative forcing (PCRs)

RCP2.6; RCP4.5; RCP6.0; RCP8.5



Policy innervation

Carbon tax; Carbon Constraint;
Nation Determined Commitments (NDC)



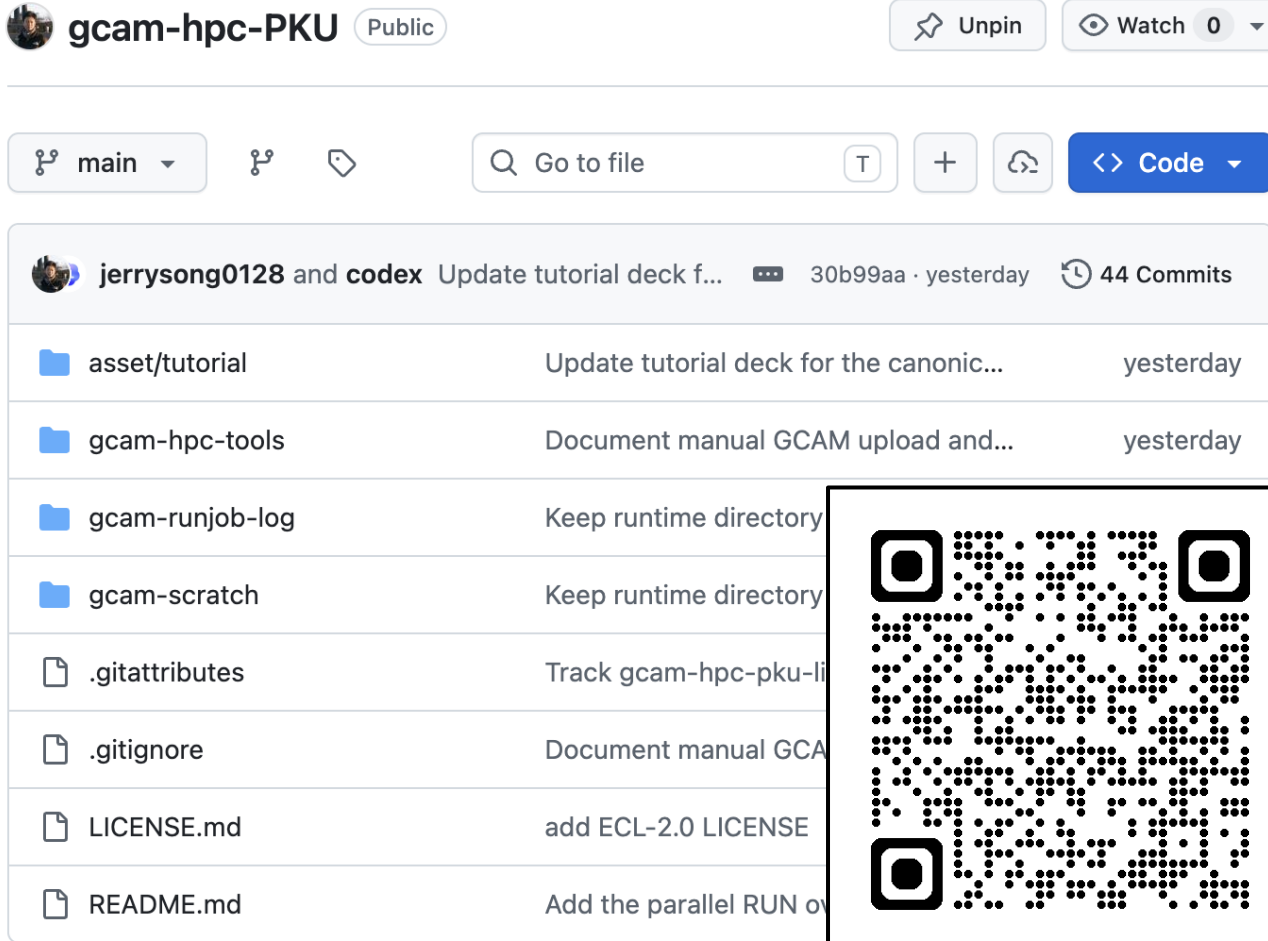
Technological potential

High CCS; High Nuclear; High Renewables;

Thousands of scenarios



GCAM with High-Performance Computing (HPC) Cluster

The image shows a GitHub repository page for 'gcam-hpc-PKU'. At the top, it indicates the repository is 'Public' and has 0 watches. Below the repository name, there are tabs for 'main' and a search bar. A list of files and folders is displayed, including 'asset/tutorial', 'gcam-hpc-tools', 'gcam-runjob-log', 'gcam-scratch', '.gitattributes', '.gitignore', 'LICENSE.md', and 'README.md'. A QR code is overlaid on the right side of the file list.

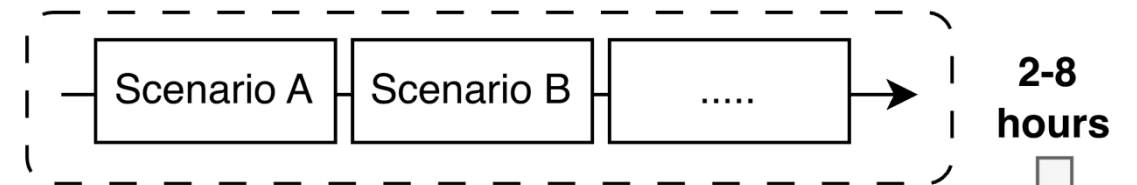
File/Folder	Description	Last Commit
asset/tutorial	Update tutorial deck for the canonic...	yesterday
gcam-hpc-tools	Document manual GCAM upload and...	yesterday
gcam-runjob-log	Keep runtime directory	
gcam-scratch	Keep runtime directory	
.gitattributes	Track gcam-hpc-pku-li	
.gitignore	Document manual GCA	
LICENSE.md	add ECL-2.0 LICENSE	
README.md	Add the parallel RUN ov	

<https://github.com/jerrysong0128/gcam-hpc-PKU>

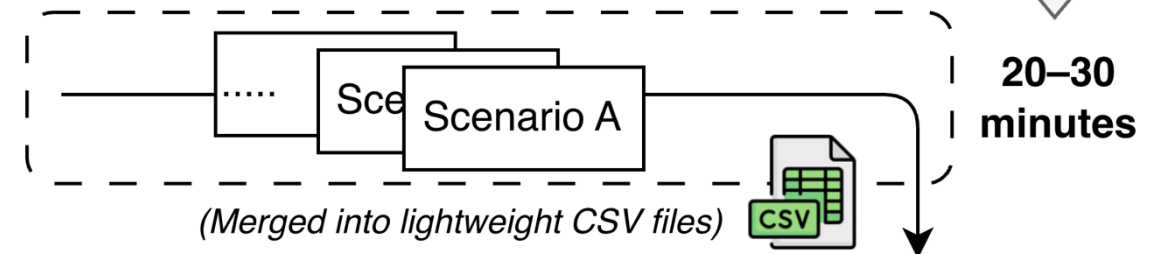
Parallel Runs for Scenario Matrix Construction

(Compatible with any version of GCAM-Core/GCAM-China)

Sequential Execution:

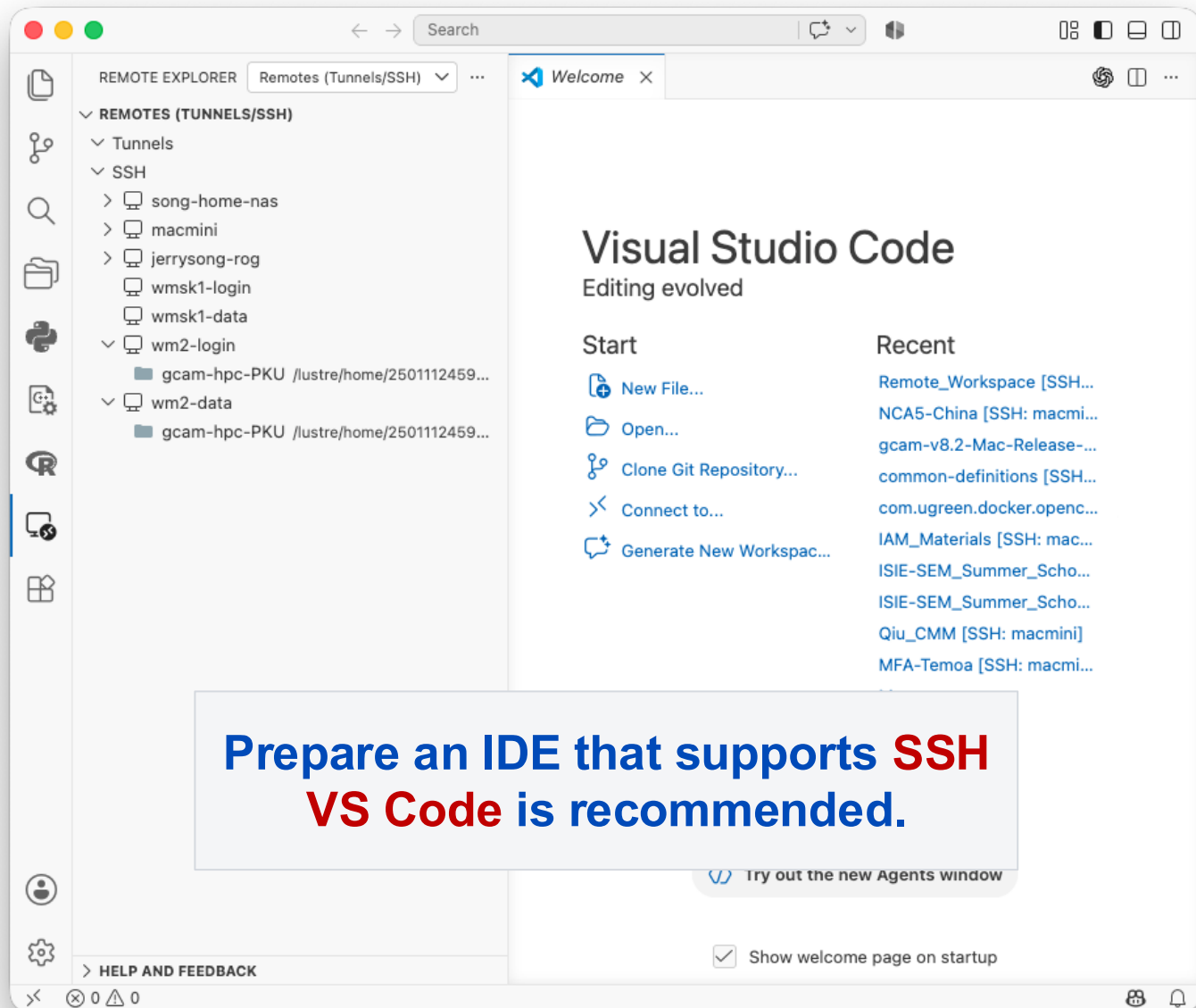


High-Performance Parallel Execution:



Use HPC to model large scenario ensemble to **SAVE TIME**

0--IDE: VS Code&Connect to HPC

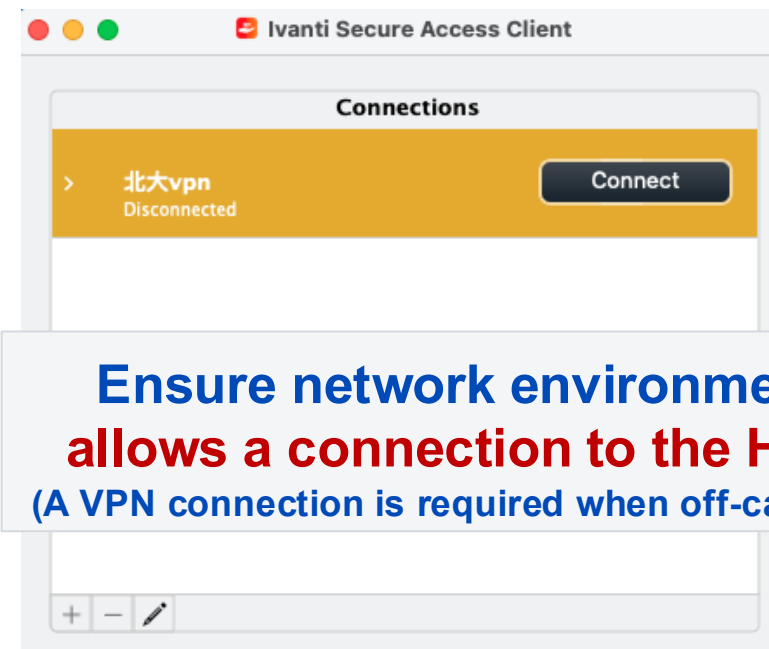


https://its.pku.edu.cn/service_1_vpn.jsp



VPN 服务

为方便北京大学校园网用户在校外（家中、出差或国外）访问校园网资源，计算中心提供VPN服务，可安全地接入校园网，如同在校内一样方便地访问学校全部的内网资源与服务（如校内门户、电子期刊数据资源等）。



0--gcam-hpc-PKU repo structure

All scripts are organized according to the following directory structure;

```
gcam-hpc-PKU/
├─ gcam-core/ 或 gcam-china-*/      GCAM source code
├─ gcam-hpc-tools/
│   ├─ build-tools/                environment.sh · build dep · profiles
│   ├─ run-tools/                  RUN · QUERY · Merge CSV
│   └─ query-tools/                QUERY generator · user QUERY XML
│   └─ local-tools/                local CSV tools
│   └─ configuration-sets/          configuration batch XML
│   └─ run-pipeline.sh              sync → expand scenario → Slurm → Merge
├─ gcam-runjob-log/                Slurm log
└─ gcam-scratch/                   run input · output
```

1-BUILD

Use the GCAM source code to compile the Linux version of `gcam.exe`; extract the HPC dependencies and load `environment.sh` before compiling.

2-RUN

`run-pipeline.sh` synchronizes inputs with pre-configured queries to generate scenarios and Slurm jobs; the same set of queries is executed after each scenario runs.

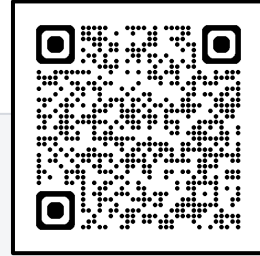
3-QUERY

XML queries must be prepared in advance; CSV results from all query scenarios are ultimately merged into a single file.

1--BUILD: clone repo and prepare to build

1 · Clone Repo

```
git clone https://github.com/jerrysong0128/gcam-hpc-PKU.git
cd gcam-hpc-PKU
```



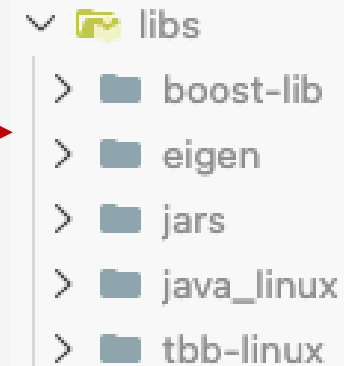
2 · Download the GCAM source code locally and upload it to the HPC

```
gcam-core: github.com/JGCRI/gcam-core/releases
gcam-china: github.com/umd-cgs/gcam-china/releases
# Source code must include cvs/objects/build/linux.
```

3 · Extract dependencies and load the environment.

```
cd gcam-hpc-tools/build-tools
unzip gcam-hpc-pku-libraries.zip
unzip archive/libraries.zip          # Create libs/

export GCAM_HPC_WORKSPACE=/absolute/path/to/gcam-hpc-PKU
export GCAM_HPC_CLUSTER=wm2
source environment.sh
```

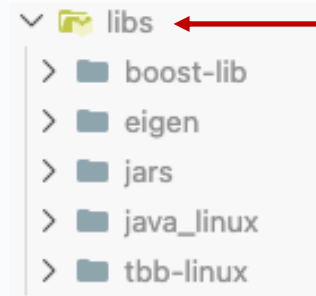


environment.sh automatically detects the gcam-core / gcam-china source code directory and loads profiles/<cluster>.profile.

1--BUILD: build gcam.exe

Dependencies folder

```
cd gcam-hpc-tools/build-tools
unzip gcam-hpc-pku-libraries.zip
unzip archive/libs.zip
```



Refer to `wm2.profile` and modify the paths for the corresponding HPC system to point to the prepared libraries.

Load the environment and compile

```
export GCAM_HPC_WORKSPACE=/path/to/gcam-hpc-PKU
export GCAM_HPC_CLUSTER=wm2
source gcam-hpc-tools/build-tools/environment.sh
cd "$GCAM_LIB"
make clean
make gcam -j 16
```

Successful compilation indicator

Compilation exit code: 0 · exe/gcam.exe generated

Cluster profile build-tools/profiles/wm2.profile

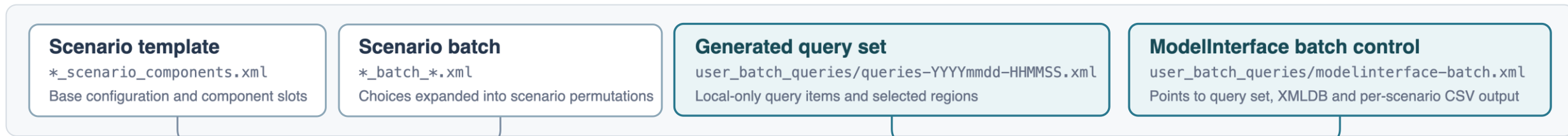
```
1 # Purpose: Define compiler, local libraries, Java, and Slurm for WEIMING-2.
2 # Author: Jingyang Song, Peking University; Jul 2026;
3 #
4 # wm2.profile -- WEIMING-2 cluster
5 # Sourced by environment.sh. Do not source directly.
6 # Provides build env (compiler, BOOST/JAVA/TBB/EIGEN, JARS) and SLURM_TEMPLATE.
7
8 module load gcc/12.2.0
9 export CXX=g++
10
11 _GCAM_LIBS="${TOOLDIR}/build-tools/libs"
12
13 export BOOST_INCLUDE="${_GCAM_LIBS}/boost-lib"
14 export BOOST_LIB="${_GCAM_LIBS}/boost-lib/lib"
15
16 export JAVA_HOME="${_GCAM_LIBS}/java_linux"
17 export JAVA_INCLUDE="${JAVA_HOME}/include"
18 export JAVA_LIB="${JAVA_HOME}/lib/server"
19 export JARS_LIB="${_GCAM_LIBS}/jars/*"
20
21 export EIGEN_INCLUDE="${_GCAM_LIBS}/eigen"
22
23 export TBB_INCLUDE="${_GCAM_LIBS}/tbb-linux/include"
24 export TBB_LIB="${_GCAM_LIBS}/tbb-linux/lib/intel64/gcc4.8"
25
26 export SLURM_TEMPLATE="${TOOLDIR}/run-tools/slurm-templates/wm2-run-template.slurm"
27
28 unset _GCAM_LIBS
29
```

Use WEIMING-2, PKU, as an example

2--RUN: Pipeline framework (optional)

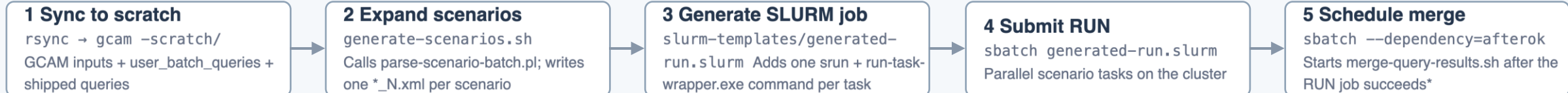
GCAM-HPC: RUN + QUERY

PRE-RUN CONFIGURATION

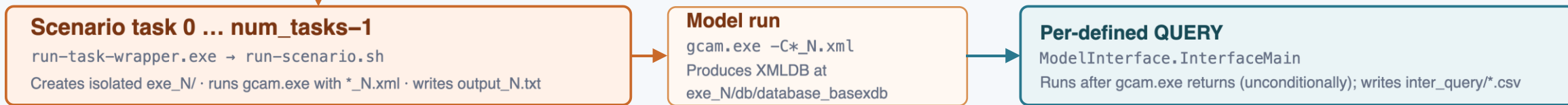


ORCHESTRATION

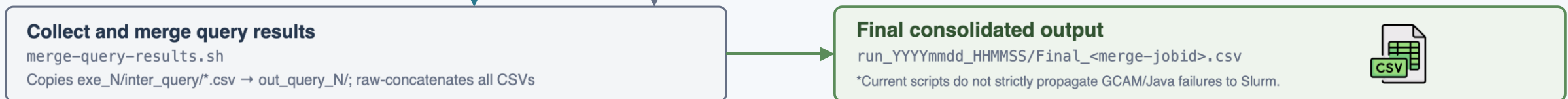
run-pipeline.sh sources build-tools/environment.sh



PARALLEL SCENARIO TASKS



AGGREGATION



2--RUN: Submit the job using run-pipeline.sh

1 · Pre-run check: QUERY configured

```
test -f gcam-hpc-tools/query-tools/user_batch_queries/modelinterface-batch.xml
# Make sure <queryFile> point to pre-defined queries-*.xml
```

2 · Submit job

```
"PATH-TO-GCAM-HPC-PKU/gcam-hpc-tools/run-pipeline.sh" \
  "PATH-TO-GCAM-HPC-PKU/gcam-hpc-tools/configuration-sets/config/v82_default/v82_default_scenario_components.xml" \
  "PATH-TO-GCAM-HPC-PKU/gcam-hpc-tools/configuration-sets/config/v82_default/v82_default_batch_2.xml" \
  Standard version settings, excluding scenario_components.
  Scenario-based batch file
```

3 · run-pipeline.sh interactive confirmation

- ① Whether to generate scenario permutations
- ② Number of tasks to run simultaneously (parallel tasks)
- ③ Whether to submit immediately using sbatch

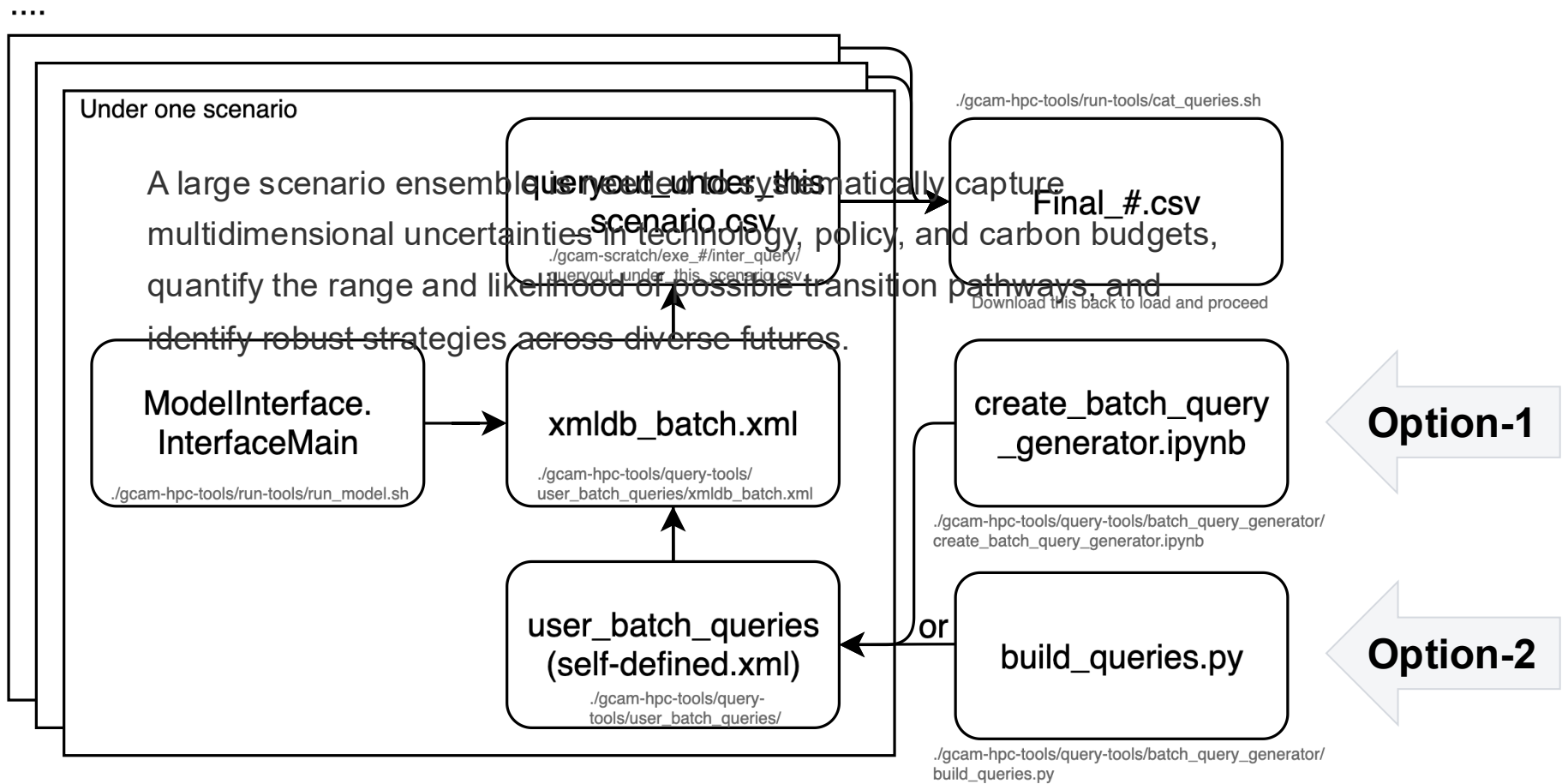
4 · Check operating status

```
squeue -u $(whoami)
bash gcam-hpc-tools/run-tools/monitor-jobs.sh
```

3--QUERY: Pre-defined Query XML

1 · Generate bath user queries

Generate the query XML before submitting the RUN, and point `xmldb\_batch.xml` to it. At runtime, after `gcam.exe` finishes, each `run\_model.sh` task invokes `ModelInterface` to execute this set of pre-configured queries and export the results to CSV.



3--QUERY: Pre-defined Query XML

1 · Generate bath user queries (Option-1)

query-tools/batch_query_generator/interactive-query-builder.ipynb # Notebook

The screenshot shows a Jupyter Notebook cell with the following Python code:

```
regions = create_batch_query_tools.read_regions_from_csv(gcam_core_path+"/input/gcamdata/inst/extdata/common/GCAM_region_names.csv")
queries = gcamreader.parse_batch_query(gcam_core_path+"/output/queries/Main_queries.xml")
result = create_batch_query_tools.select_queries_and_regions(queries, regions)# start selection panel
```

Below the code, the execution output shows a successful run in 0.1s. The interactive query builder UI is displayed with two panels:

- Regions:** A list of regions including All, USA, Africa_Eastern, Africa_Northern, Africa_Southern, Africa_Western, Australia_NZ, Brazil, Canada, Central America and Caribbean, Central Asia, China, EU-12, EU-15, and Ukraine.
- Queries:** A list of queries including All, primary energy consumption by region (avg fossil efficiency), primary energy consumption by region (direct equivalent), primary energy consumption with CCS by region (direct equivalent), resource production, resource production by tech and vintage, resource supply curves, regional primary energy prices, elec gen by region (incl CHP), elec gen by subsector, elec gen by gen tech, elec gen by gen tech and cooling tech, elec gen by gen tech and cooling tech and vintage, elec gen by gen tech and cooling tech (new), and elec energy input by subsector.

A green button labeled "Confirm Selection" is located at the bottom left of the UI.

交互式 QUERY 生成器

2 · config modelinterface-batch.xml

run-pipeline.sh synchronizes user_batch_queries/ to scratch/output/xmlldb_queries/; <queryFile> must point to the synchronized file.

```
<queryFile>../output/xmlldb_queries/queries-20260726-110822.xml</queryFile>
```

3--QUERY: Pre-defined Query XML

1 · Generate bath user queries (Option-2)

```
query-tools/batch_query_generator/query-config.yaml #Query XML settings
```

```
# GCAM batch query builder config
#
# How to use:
# 1. Copy region names from available_regions into selected_regions.
# 2. Copy query titles from available_queries into selected_queries.
# 3. Run: python build_queries.py query-config.yaml
#
# Use All to select every available region or query.
```

```
selected_regions:
```

```
- All
```

```
selected_queries:
```

```
- iron and steel production by region
- iron and steel production by tech
```

```
query-tools/batch_query_generator/build_queries.py #Create Query XML with script
```

```
• (base) jerrysong@bogon Remote_Workspace % cd /Users/jerrysong/Study_PKU/Projects/Remote_Workspace/gcam-hpc-PKU/gcam-hpc-tools/query-tools/batch_query_generator
• (base) jerrysong@bogon batch_query_generator % python build_queries.py query-config.yaml
Generated: /Users/jerrysong/Study_PKU/Projects/Remote_Workspace/gcam-hpc-PKU/gcam-hpc-tools/query-tools/user_batch_queries/queries-20260727-220824.xml
Queries: 2
Regions: 32
```

2 · config modelinterface-batch.xml

run-pipeline.sh 将 user_batch_queries/ 同步到 scratch/output/xmldb_queries/; <queryFile> 必须指向同步后的文件。

```
<queryFile>../output/xmldb_queries/queries-20260726-110822.xml</queryFile>
```

4--OUTPUT: CSV & logs

Output folder

```
gcam-runjob-log/  
├─ gcamCatCSV.<jobid>.out      # job log for Pipeline merge  
└─ gcamRunjob.<jobid>.out      # job log for Pipeline run
```

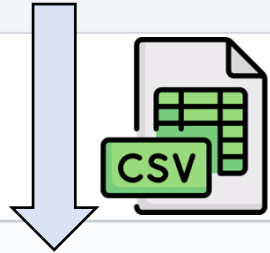
```
gcam-scratch/output/run_YYYYmmdd_HHMMSS/  
├─ output_<task>.txt          # gcam log per scenario  
├─ out_query_<task>/          # queried CSV per scenario  
└─ Final_<merge-jobid>.csv    # merged CSV
```

run-pipeline.sh related error

gcam related error

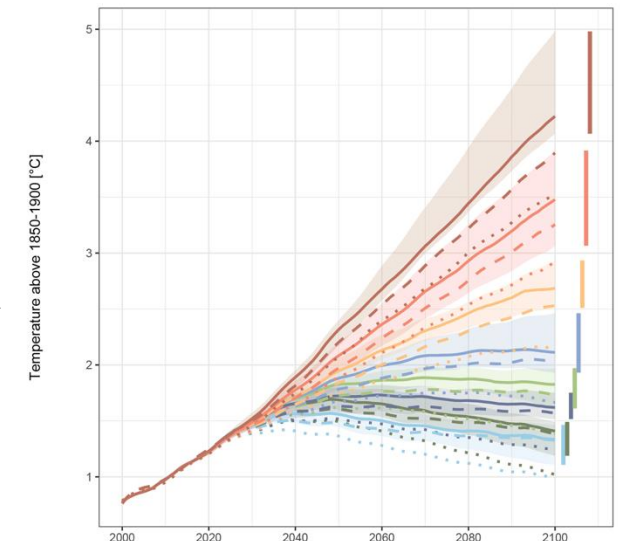
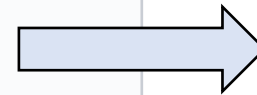
single gcam output

All scenario gcam output

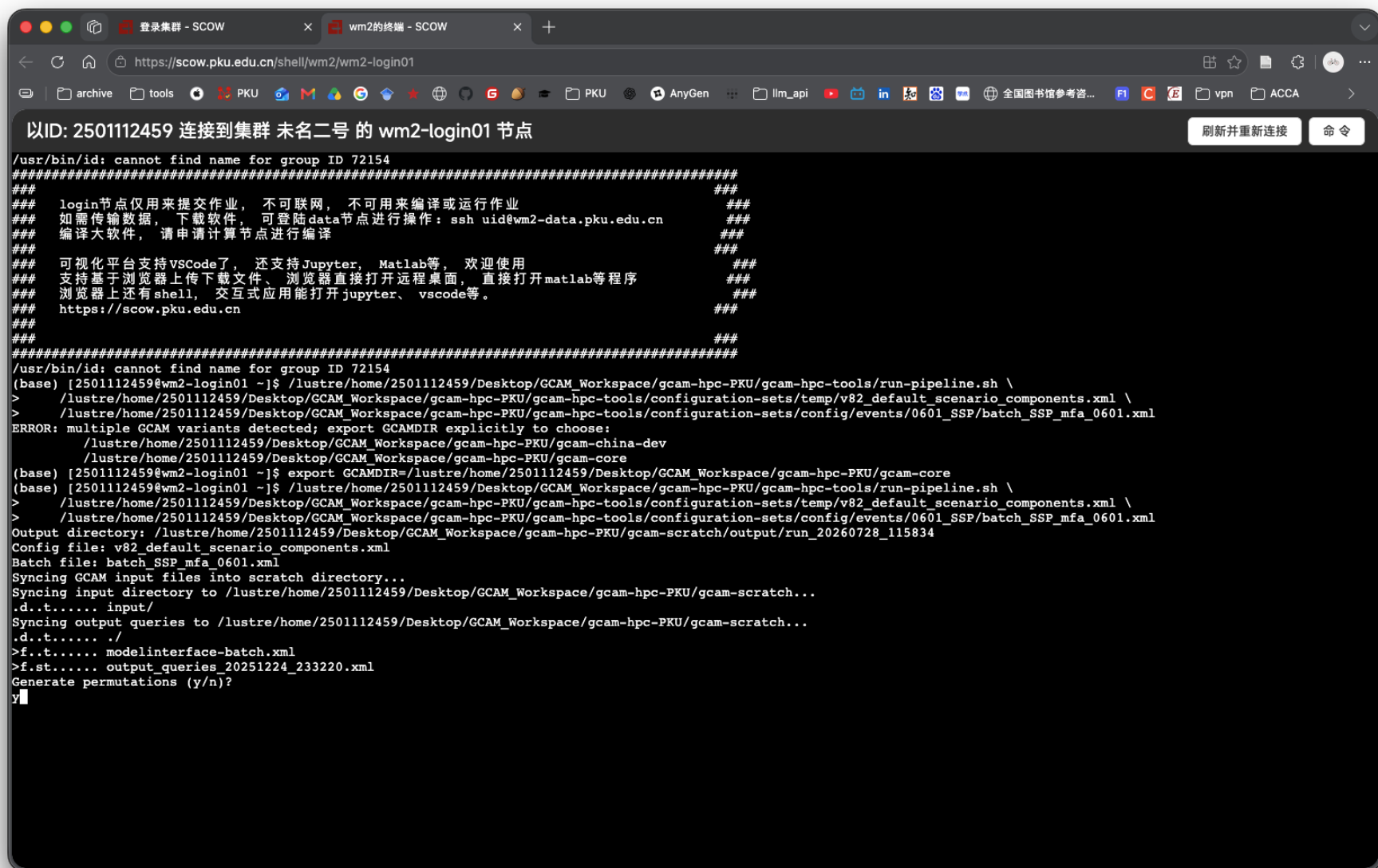


**Perform
post-processing
locally**

```
gcam-hpc-PKU/gcam-hpc-tools/local-tools/  
├─ data_prepare_main_HPC.R  
├─ diag_header_HPC.R  
└─ diag_parser_HPC.R
```



2--RUN: Demonstration

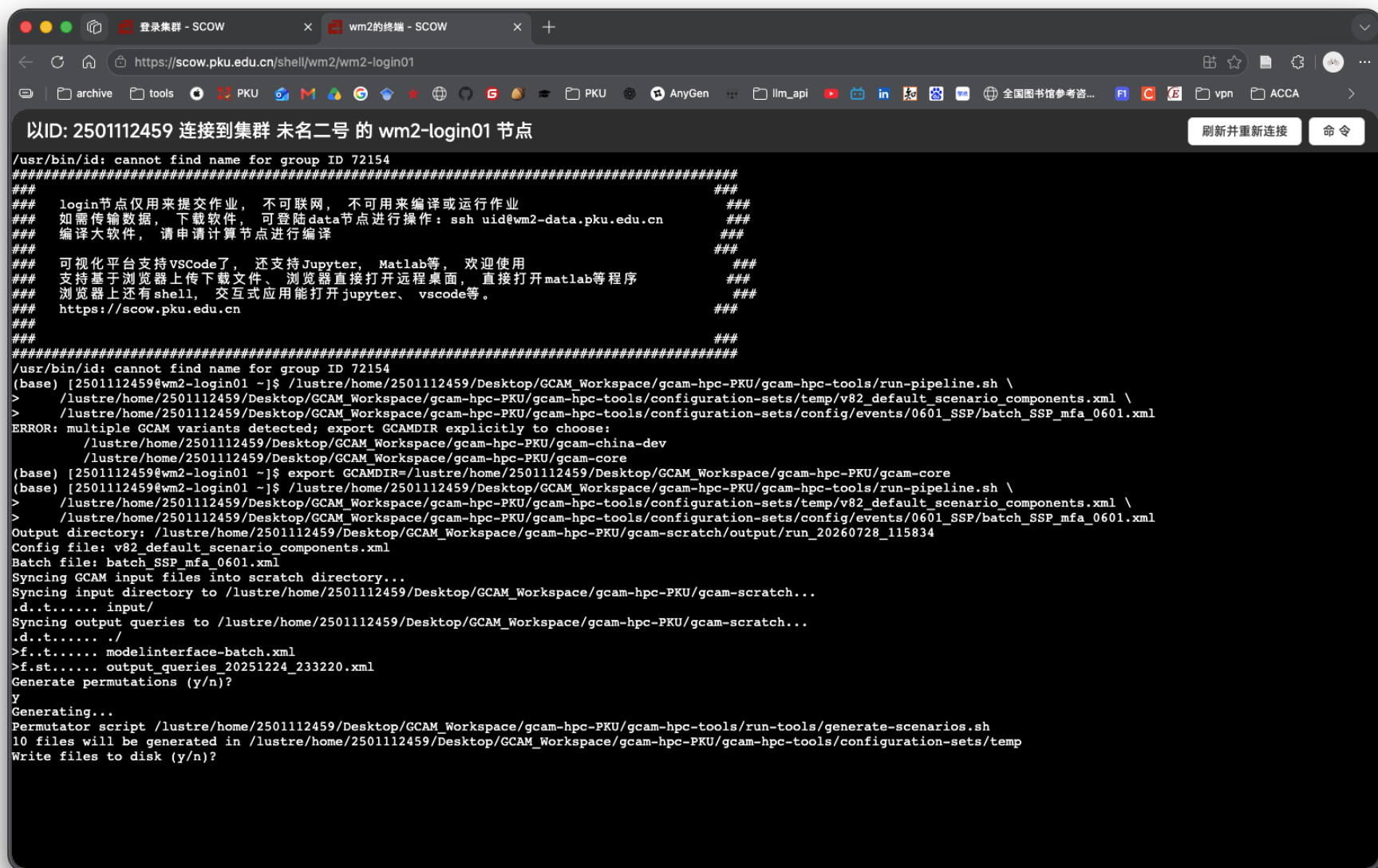


The screenshot shows a web browser window with a terminal interface. The browser's address bar displays `https://scow.pku.edu.cn/shell/wm2/wm2-login01`. The terminal window has a title bar with two tabs: "登录集群 - SCOW" and "wm2的终端 - SCOW". Below the title bar, a message reads: "以ID: 2501112459 连接到集群 未名二号的 wm2-login01 节点". To the right of this message are two buttons: "刷新并重新连接" and "命令".

The terminal output shows the following sequence of commands and responses:

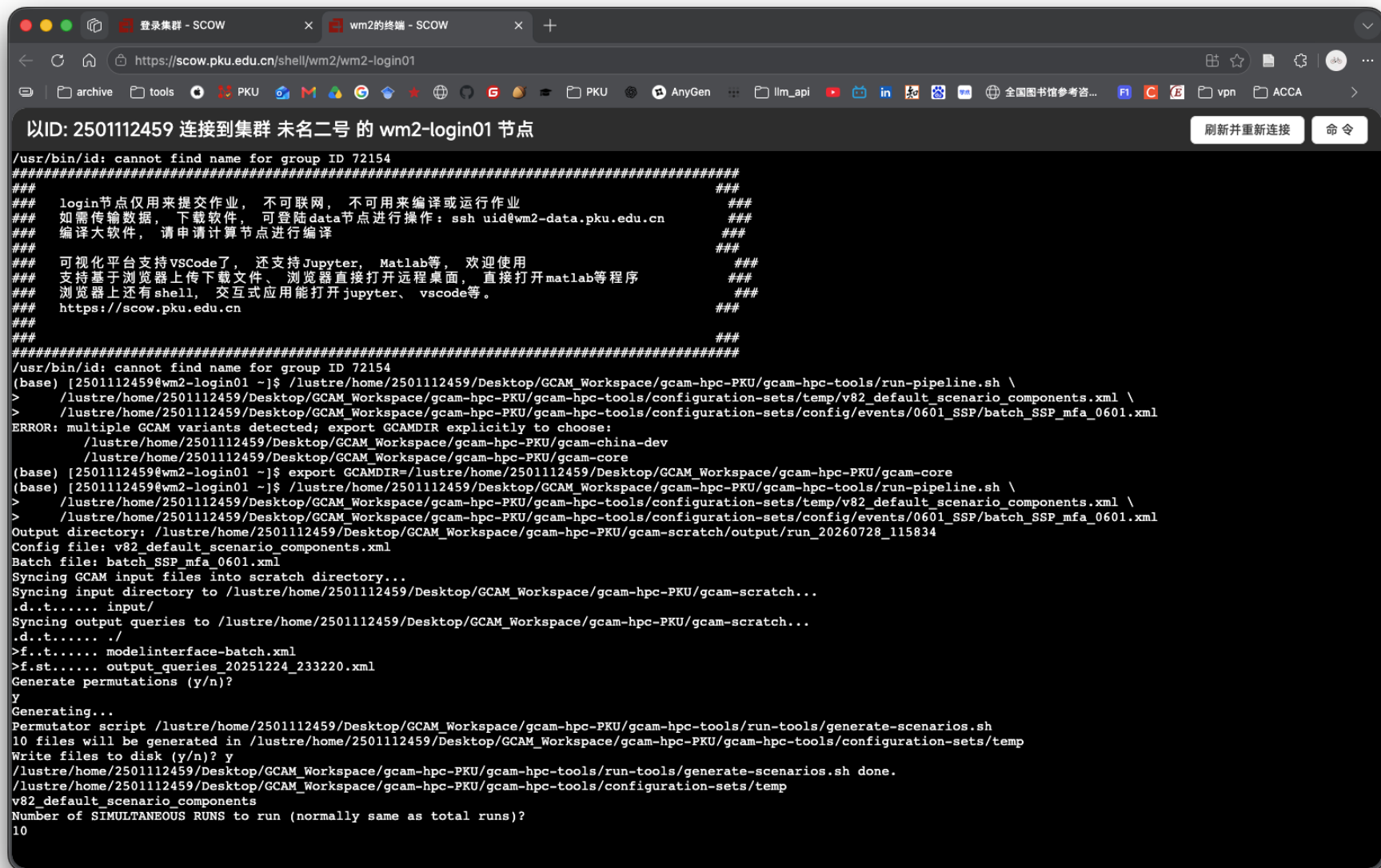
```
/usr/bin/id: cannot find name for group ID 72154
#####
###
### login节点仅用来提交作业，不可联网，不可用来编译或运行作业
### 如需传输数据，下载软件，可登陆data节点进行操作：ssh uid@wm2-data.pku.edu.cn
### 编译大软件，请申请计算节点进行编译
###
### 可视化平台支持VSCode了，还支持Jupyter，Matlab等，欢迎使用
### 支持基于浏览器上传下载文件、浏览器直接打开远程桌面，直接打开matlab等程序
### 浏览器上还有shell，交互式应用能打开jupyter、vscode等。
### https://scow.pku.edu.cn
###
#####
/usr/bin/id: cannot find name for group ID 72154
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-china-dev
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
Syncing GCAM input files into scratch directory...
Syncing input directory to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d..t..... input/
Syncing output queries to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d..t..... ./
>f..t..... modelinterface-batch.xml
>f.st..... output_queries_20251224_233220.xml
Generate permutations (y/n)?
y
```

2--RUN: Demonstration



```
#####
###
### login节点仅用来提交作业，不可联网，不可用来编译或运行作业
### 如需传输数据，下载软件，可登陆data节点进行操作：ssh uid@wm2-data.pku.edu.cn
### 编译大软件，请申请计算节点进行编译
###
### 可视化平台支持VSCode了，还支持Jupyter，Matlab等，欢迎使用
### 支持基于浏览器上传下载文件、浏览器直接打开远程桌面，直接打开matlab等程序
### 浏览器上还有shell，交互式应用能打开jupyter、vscode等。
### https://scow.pku.edu.cn
###
#####
/usr/bin/id: cannot find name for group ID 72154
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-china-dev
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
Syncing GCAM input files into scratch directory...
Syncing input directory to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d.t..... input/
Syncing output queries to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d.t..... ./
>f.t..... modelinterface-batch.xml
>f.st..... output_queries_20251224_233220.xml
Generate permutations (y/n)?
y
Generating...
Permutator script /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh
10 files will be generated in /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
Write files to disk (y/n)?
```

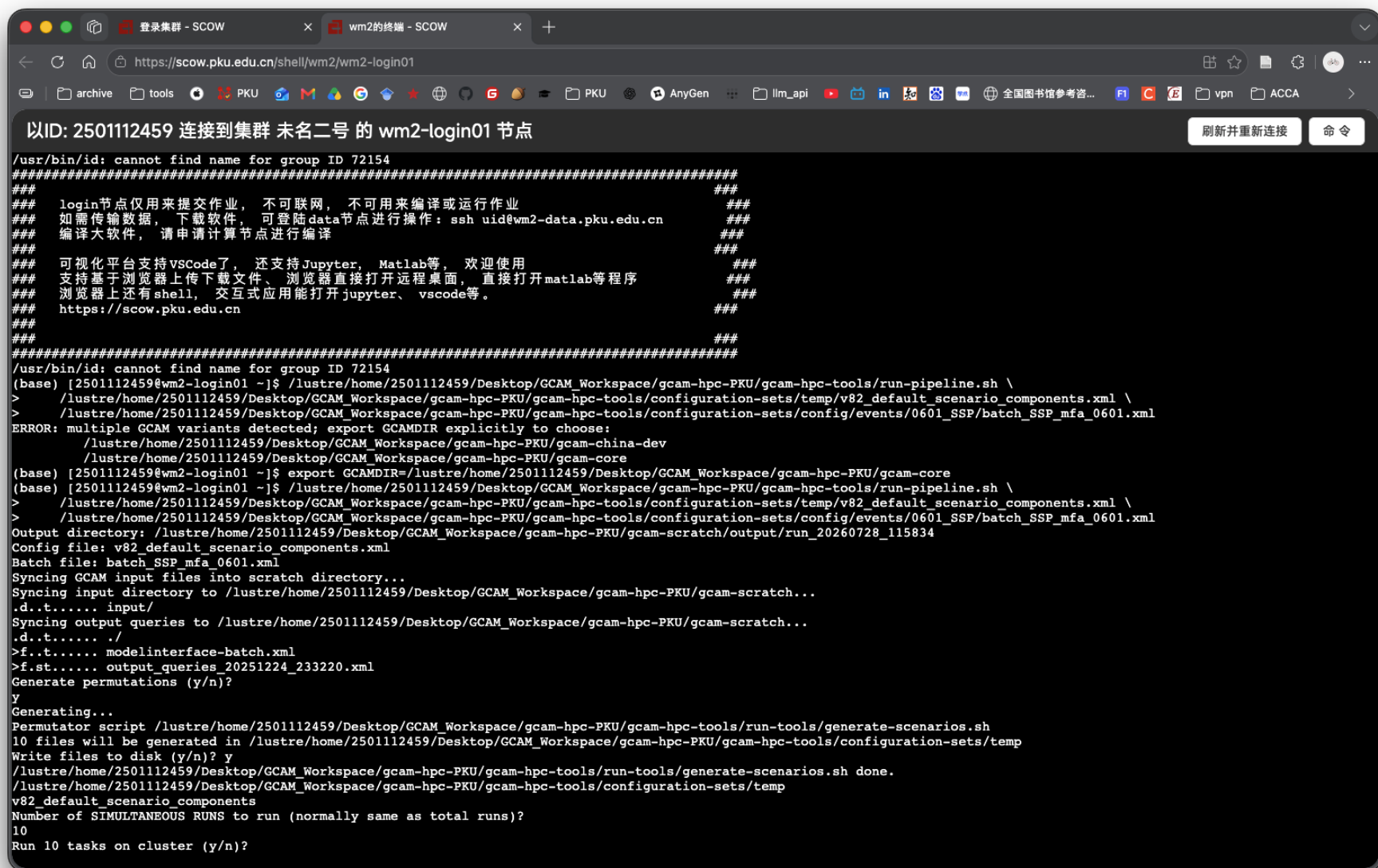
2--RUN: Demonstration



```
以ID: 2501112459 连接到集群 未名二号的 wm2-login01 节点
刷新并重新连接 命令

/usr/bin/id: cannot find name for group ID 72154
#####
###
### login节点仅用来提交作业，不可联网，不可用来编译或运行作业
### 如需传输数据，下载软件，可登陆data节点进行操作：ssh uid@wm2-data.pku.edu.cn
### 编译大软件，请申请计算节点进行编译
###
### 可视化平台支持VSCode了，还支持Jupyter，Matlab等，欢迎使用
### 支持基于浏览器上传下载文件、浏览器直接打开远程桌面，直接打开matlab等程序
### 浏览器上还有shell，交互式应用能打开jupyter、vscode等。
### https://scow.pku.edu.cn
###
#####
/usr/bin/id: cannot find name for group ID 72154
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-china-dev
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
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Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
Syncing GCAM input files into scratch directory...
Syncing input directory to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d.t..... input/
Syncing output queries to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d.t..... ./
>f.t..... modelinterface-batch.xml
>f.st..... output_queries_20251224_233220.xml
Generate permutations (y/n)?
y
Generating...
Permutator script /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh
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Write files to disk (y/n)? y
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v82_default_scenario_components
Number of SIMULTANEOUS RUNS to run (normally same as total runs)?
10
```

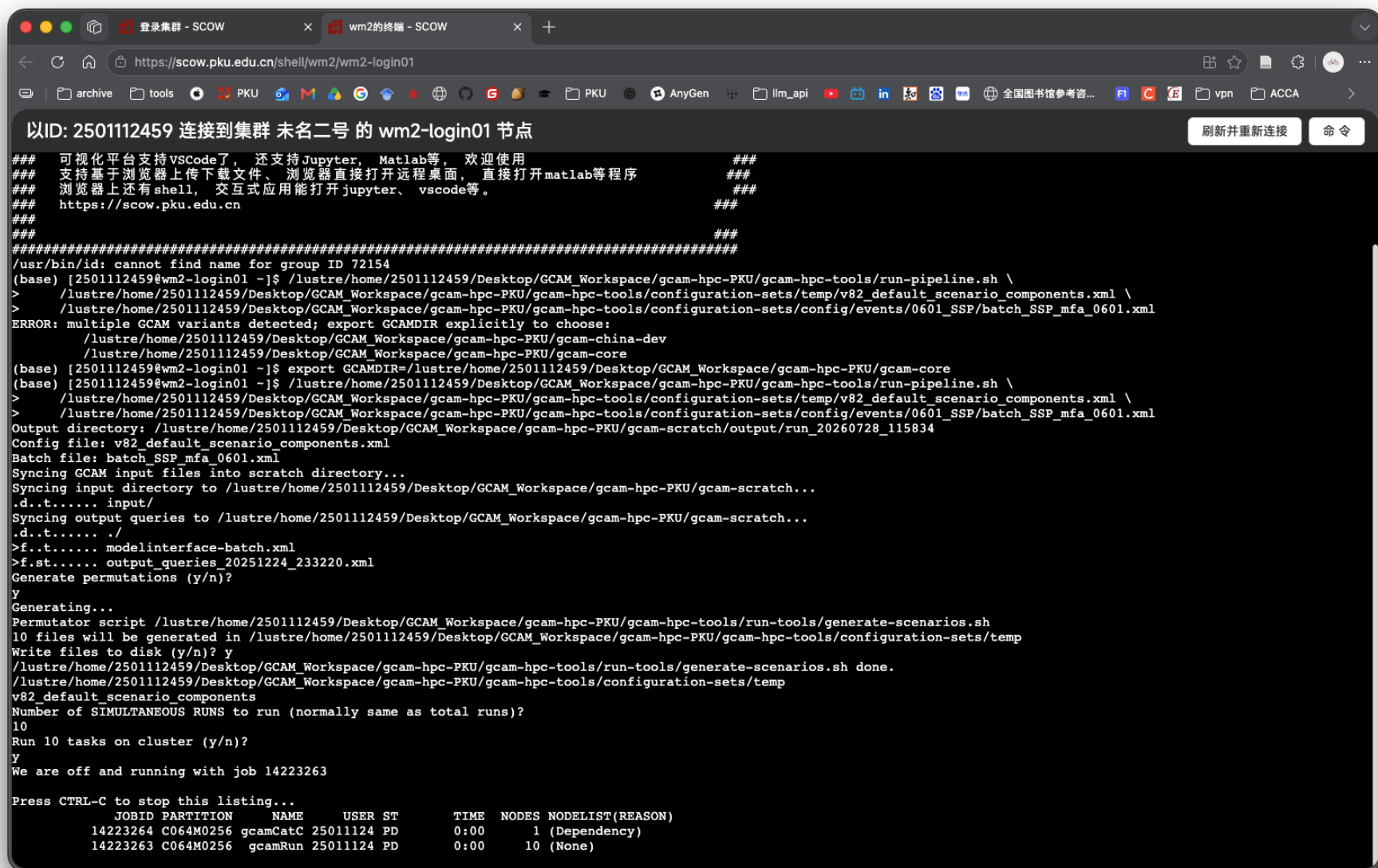
2--RUN: Demonstration



```
以ID: 2501112459 连接到集群 未名二号的 wm2-login01 节点 刷新并重新连接 命令

/usr/bin/id: cannot find name for group ID 72154
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v82_default_scenario_components
Number of SIMULTANEOUS RUNS to run (normally same as total runs)?
10
Run 10 tasks on cluster (y/n)?
```

2--RUN: Demonstration



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以ID: 2501112459 连接到集群 未名二号的 wm2-login01 节点
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/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh done.
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
v82_default_scenario_components
Number of SIMULTANEOUS RUNS to run (normally same as total runs)?
10
Run 10 tasks on cluster (y/n)?
y
We are off and running with job 14223263

Press CTRL-C to stop this listing...
  JOBID PARTITION   NAME      USER ST   TIME  NODES MODELIST(REASON)
    14223264 C064M0256 gcamCatC 25011124 PD   0:00     1 (Dependency)
    14223263 C064M0256 gcamRun  25011124 PD   0:00    10 (None)
```

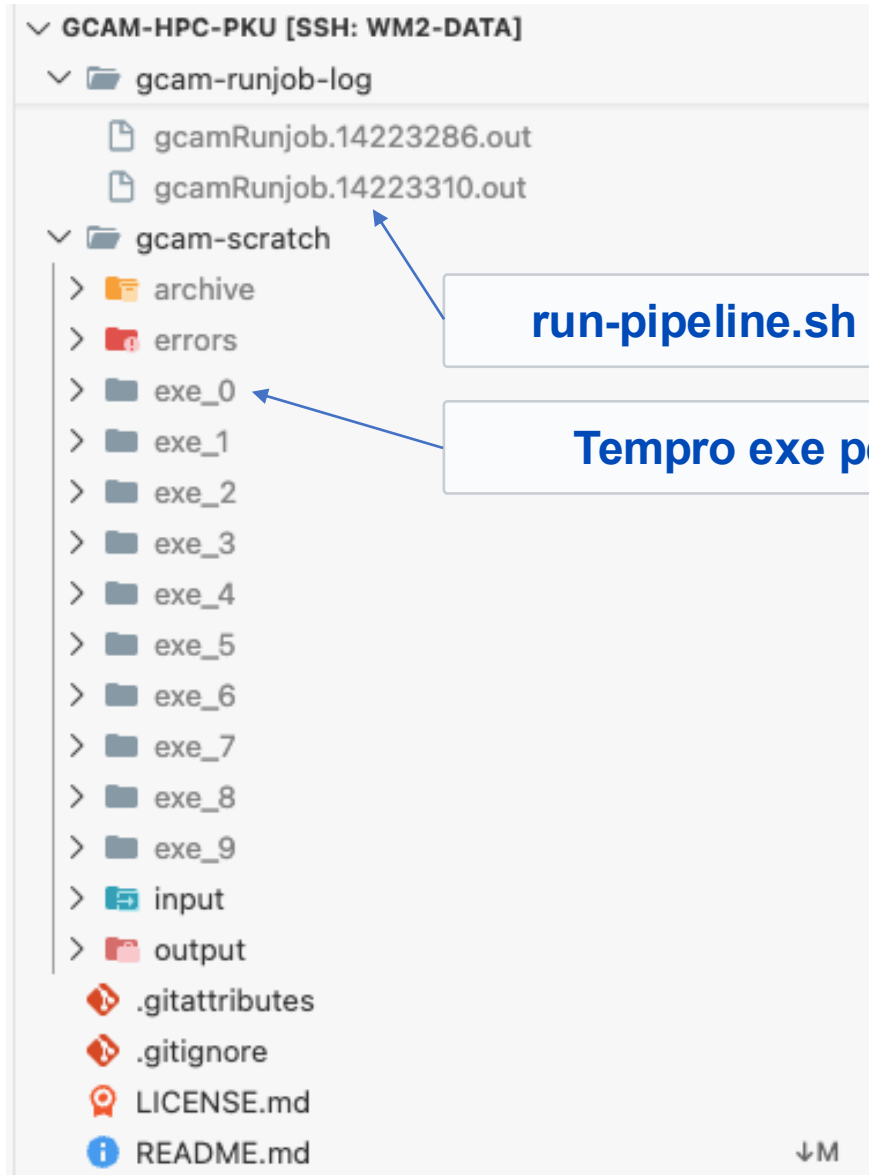

2--RUN: Demonstration

The screenshot displays the SCOW web interface in a browser window. The address bar shows the URL `https://scow.pku.edu.cn/jobs/runningJobs`. The page title is "未结束的作业 - SCOW". The left sidebar contains navigation links: "未结束的作业" (selected), "所有作业", and "提交作业". The main content area is titled "未结束的作业" and features a search bar with "集群: 未名二号" and "作业ID:" fields, along with "搜索" and "刷新" buttons. Below the search bar is a table of running jobs.

作业ID	作业名	账户	分区	QOS	节点数	CPU核心数	加速卡卡数	状态	操作
14223311	gcamCatCSV	hpc2306191132	C064M0256G	normal	1	8	0	PENDING	
14223310	gcamRun	hpc2306191132	C064M0256G	normal	10	80	0	RUNNING	

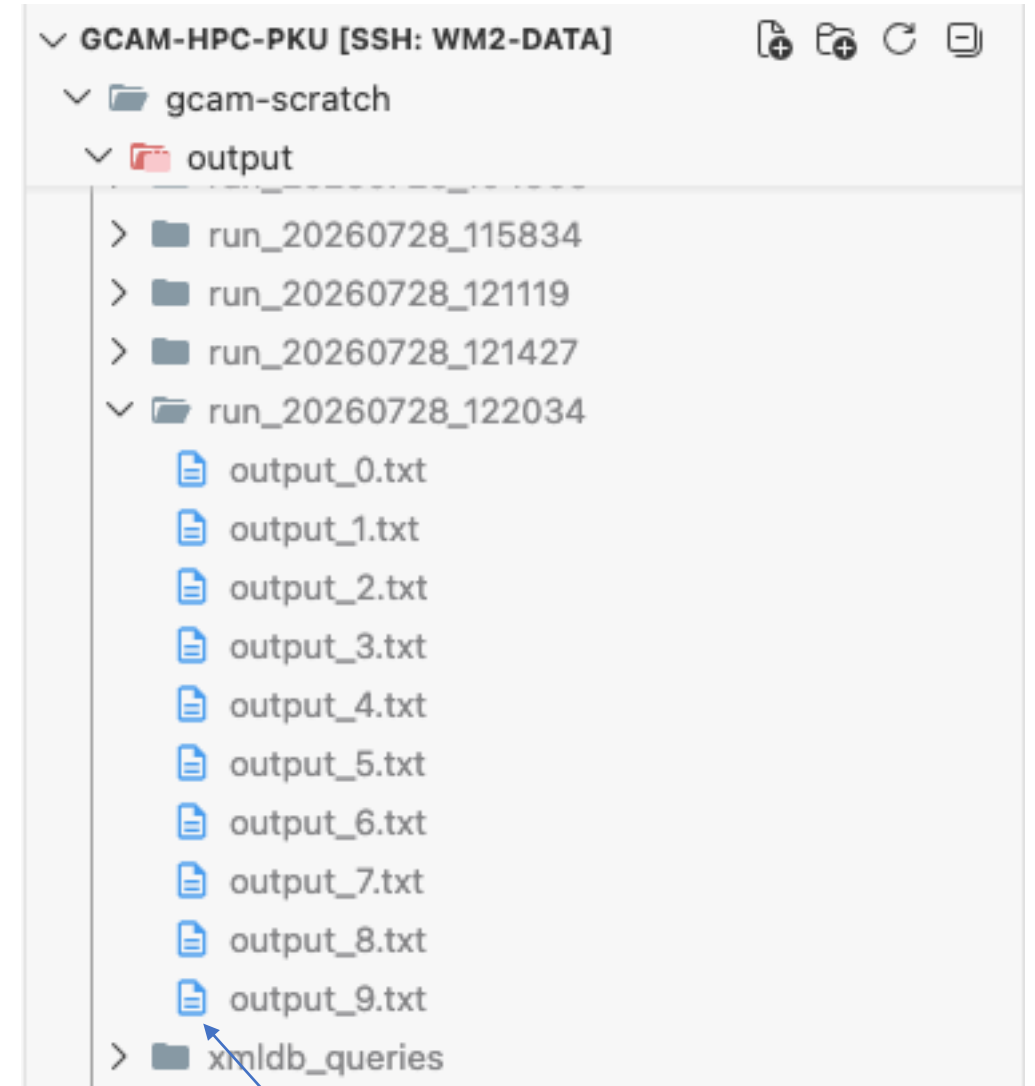
At the bottom right of the table, there is a pagination control showing "< 1 >" and "50 条/页".

2--RUN: Demonstration



run-pipeline.sh related error

Tempore exe per scenario



gcam related error

More things you need to know

Common Issues

- Failure to extract archive/libs.zip from gcam-hpc-pku-libs.zip, resulting in an incomplete build-tools/libs/ directory
- Multiple gcam-* source directories in the workspace, or an incorrect GCAMDIR path
- Incorrect cluster profile, or mismatched Slurm partition, modules, or dependency paths
- queries-*.xml files not generated before RUN, or modelinterface-batch.xml still referencing outdated filenames
- Region or query names in QUERY do not match those defined in the configuration files

Tools and Documentation

Repository <https://github.com/jerrysong0128/gcam-hpc-PKU>

Wiki Home · 1-Build · 2-Run · 3-Query

Source Code gcam-core: JGCRI/gcam-core · gcam-china: umd-cgs/gcam-china

Acknowledge repo is built on previous work from Jay Fuhrman, University of Virginia <https://github.com/jayfuhrman/gcam-hpc-starterpack>

Three-Step Quick Start

BUILD Upload source code → Extract libraries → source environment.sh → make gcam

QUERY Select regions and queries → build_queries.py → Configure modelinterface-batch.xml

RUN run-pipeline.sh → Run GCAM + QUERY for each scenario → Automatically merge CSV files

For further details, please refer to the Wiki page

Home
Jerry Song edited this page yesterday · [7 revisions](#)

gcam-hpc-PKU Wiki

Parallel Runs for Scenario Matrix Construction

(Compatible with any version of GCAM-Core/GCAM-China)

Sequential Execution:

Scenario A → Scenario B → → 2-8 hours

High-Performance Parallel Execution:

..... → Scenario A → (Merged into lightweight CSV files) → 20-30 minutes

本 Wiki 说明如何在北京大学高性能计算平台上完成 GCAM 的 BUILD → QUERY 配置 → RUN → CSV 合并流程。

English: [Home-EN](#) | [Build-EN](#) | [Run-EN](#) | [Query-EN](#)

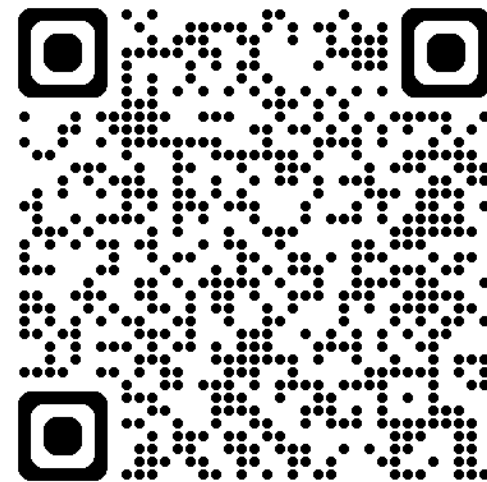
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 - [首页](#)
 - [1. 编译 GCAM](#)
 - [2. 运行 GCAM](#)
 - [3. 查询输出](#)
- English
 - [Home](#)
 - [1. Build GCAM](#)
 - [2. Run GCAM](#)
 - [3. Query GCAM Output](#)

Clone this wiki locally

<https://github.com/jerrysong0128/>

<https://github.com/jerrysong0128/gcam-hpc-PKU/wiki>



We build our repo on
previous work from
Jay Fuhrman,
University of Virginia
<https://github.com/jayfuhrman/gcam-hpc-starterpack>

GCAM-HPC-PKU

基于计算集群的GCAM工具

Jerry Song 宋京阳
jerrysong2025@stu.pku.edu.cn
2026-8-1

